

40 Number Programs with Examples and Brief Explanation

1. Sum of Digits

Explanation: Adds all digits of a number to get a single total.

Example Output: Input: 123 → Output: 6

2. Reverse of a Number

Explanation: Reverses the order of digits in a number.

Example Output: Input: 123 → Output: 321

3. Palindrome Number

Explanation: Checks whether a number reads the same forward and backward.

Example Output: 121 → Palindrome

4. Digital Root

Explanation: Repeatedly adds digits until a single digit is obtained.

Example Output: 987 → 6

5. Count Digits

Explanation: Counts total digits present in a number.

Example Output: 4589 → 4

6. Product of Digits

Explanation: Multiplies all digits of a number.

Example Output: 234 → 24

7. Armstrong Number

Explanation: Checks if sum of powers of digits equals the number.

Example Output: 153 → Armstrong

8. Strong Number

Explanation: Checks if sum of factorials of digits equals the number.

Example Output: 145 → Strong

9. Spy Number

Explanation: Checks if sum of digits equals product of digits.

Example Output: 1124 → Spy

10. Perfect Number

Explanation: Checks if sum of proper divisors equals the number.

Example Output: 28 → Perfect

11. Automorphic Number

Explanation: Checks if square of number ends with the number.

Example Output: 25 → Automorphic

12. Neon Number

Explanation: Checks if sum of digits of square equals the number.

Example Output: 9 → Neon

13. Duck Number

Explanation: Checks if number contains zero (not at start).

Example Output: 1023 → Duck

14. Harshad Number

Explanation: Checks if number is divisible by sum of digits.

Example Output: 18 → Harshad

15. Swap First & Last Digit

Explanation: Interchanges the first and last digits of a number.

Example Output: 1234 → 4231

16. Largest Digit

Explanation: Finds the largest digit in a number.

Example Output: 5729 → 9

17. Smallest Digit

Explanation: Finds the smallest digit in a number.

Example Output: 5729 → 2

18. Remove Zero Digits

Explanation: Removes all zeros from the number.

Example Output: 102030 → 123

19. Prime Number

Explanation: Checks if a number has exactly two factors.

Example Output: 7 → Prime

20. Prime Digits Only

Explanation: Checks if all digits are prime digits.

Example Output: 235 → All Prime

21. Count Even Digits

Explanation: Counts how many even digits are present.

Example Output: 2468 → 4

22. Count Odd Digits

Explanation: Counts how many odd digits are present.

Example Output: 1357 → 4

23. Sum of Even Digits

Explanation: Adds only even digits of the number.

Example Output: 246 → 12

24. Sum of Odd Digits

Explanation: Adds only odd digits of the number.

Example Output: 135 → 9

25. Fibonacci Check

Explanation: Checks if number belongs to Fibonacci series.

Example Output: 13 → Fibonacci

26. Decimal to Binary

Explanation: Converts decimal number into binary form.

Example Output: 10 → 1010

27. Binary to Decimal

Explanation: Converts binary number into decimal form.

Example Output: 1010 → 10

28. GCD of Two Numbers

Explanation: Finds greatest common divisor of two numbers.

Example Output: 12,18 → 6

29. LCM of Two Numbers

Explanation: Finds least common multiple of two numbers.

Example Output: 12,18 → 36

30. Count Factors

Explanation: Counts total factors of a number.

Example Output: 12 → 6

31. Sum of Factors

Explanation: Finds sum of all factors except the number itself.

Example Output: 6 → 6

32. Buzz Number

Explanation: Checks if number is divisible by 7 or ends with 7.

Example Output: 7 → Buzz

33. Happy Number

Explanation: Checks if repeated sum of squares of digits equals 1.

Example Output: 19 → Happy

34. Sunny Number

Explanation: Checks if next number is a perfect square.

Example Output: 8 → Sunny

35. Disarium Number

Explanation: Checks if sum of digits powered by position equals number.

Example Output: 135 → Disarium

36. Peterson Number

Explanation: Checks if sum of factorial of digits equals number.

Example Output: 145 → Peterson

37. Factorial

Explanation: Calculates factorial of a number.

Example Output: 5! → 120

38. Fibonacci Series

Explanation: Generates Fibonacci series up to N terms.

Example Output: 0 1 1 2 3

39. Power of Number

Explanation: Calculates power without built-in functions.

Example Output: 2^3 → 8

40. Magic Number

Explanation: Checks if digital root of number is 1.

Example Output: 19 → Magic